

Proof by contradiction can be used to prove conditional statements. In such proofs, we first assume the negation of the conclusion. We then use the premises of the theorem and the negation of the conclusion to arrive at a contradiction. (The reason that such proofs are valid rests on the logical equivalence of $p \rightarrow q$ and $(p \wedge \neg q) \rightarrow F$.)

Let $p \rightarrow q$ be False. By defn. of conditional, p is True and q is False.

By defn. of negation, $\neg q$ is True. By defn. of AND, $p \wedge \neg q$ is True.

By defn. of conditional, $(p \wedge \neg q) \rightarrow F$ is False.

Let, $(p \wedge \neg q) \rightarrow F$ be False. By defn. of conditional, $(p \wedge \neg q)$ is True.

By defn. of AND, p is True and $\neg q$ is True. By defn. of negation, $\neg(\neg q)$ is False. By double-negation law, q is False. By defn. of conditional, $p \rightarrow q$ is False.

$$\therefore p \rightarrow q \equiv (p \wedge \neg q) \rightarrow F \quad \square$$

Note that we can rewrite a proof by contraposition of a conditional statement as a proof by contradiction. In a proof of $p \rightarrow q$ by contraposition, we assume that $\neg q$ is True. We then show that $\neg p$ must also be True. We suppose that both p and $\neg q$ are true. Then we use the steps from the proof of $\neg q \rightarrow \neg p$ to show that $\neg p$ is true. This leads to the contradiction, $p \wedge \neg p$, completing the proof.

Proofs of equivalence: To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $(p \rightarrow q)$ and $(q \rightarrow p)$ are both true. The validity of this approach is based on the tautology

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$